THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The unit of electrical energy used commercially is 1

(a) watt **(b)** joule per second **(c)** volt **(d)** kilowatt hour

2. The relation between V, I and R is 1

(a) V = I + R **(b)** V = I R **(c)** V = R / I **(d)** V = I / R

3. Nichrome is an alloy of 1

(a) nickel, chromium and iron **(b)** copper and zinc

(c) iron and carbon only **(d)** lead and tin

4. The fuse wire is made of a material of 1

(a) low melting point **(b)** zero resistance

(c) high melting point **(d)** high resistivity only

5. The symbol of a resistor is 1

(a) two parallel lines **(b)** a plain line

(c) a zig-zag line **(d)** a circle with a cross

6. Resistance of 1 m of a wire is 2 ohm. For 3 m of the same wire, the resistance is 1

(a) 6 ohm **(b)** 3 ohm **(c)** 1/3 ohm **(d)** 2 ohm

7. The formula for power is 1

(a) P = I / V **(b)** P = R / I **(c)** P = V I **(d)** P = V / I

8. A closed circuit is one in which 1

(a) no current flows **(b)** the key is open

(c) the cell is removed **(d)** current flows

9. Resistance is measured in 1

(a) watt **(b)** joule **(c)** volt **(d)** ohm

10. Heating effect of current is used in 1

(a) an electric heater **(b)** a magnet

(c) a battery **(d)** a compass

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define electric power and give its unit. 2

12. Write Ohm's law in the form of an equation. 2

13. Name two materials used in making heating elements. 2

14. What is a fuse? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the work done in moving 5 C of charge through a potential difference of 6 V. 3

16. An electric heater of 1000 W is used for 2 hours. Find the energy consumed in kWh. 3

17. How is a fuse connected in a circuit? Why? 3

OR

Write the symbols used in a circuit for a cell, a bulb, a switch (key closed) and a resistor. 3

18. Calculate the power of a device with 220 V and 2 A. 3

OR

Calculate the heat produced in 10 s by a 5 ohm resistor carrying 2 A. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Distinguish between series and parallel arrangements of resistors with respect to current, voltage, equivalent resistance, failure of one and use. 5

OR

Explain why wires of copper are used in house wiring and nichrome in heaters. 5

20. Explain the heating effect of electric current. Give the formula and two uses. 5

OR

Explain with a labelled description how to connect an ammeter and a voltmeter to measure the current through and the voltage across a resistor. 5

END OF QUESTION PAPER — Total: 40 Marks

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